Practice Quiz 07D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Construct an equivalence relation from its classes.

1. Let p, q, r, s be distinct objects and set $A = \{p, q, r, s\}$. Construct an equivalence relation R on A whose distinct equivalence classes are exactly $\{p, r\}$ and $\{q, s\}$. List all ordered pairs in R and justify that its classes are the stated sets. Separate proofs of reflexivity, symmetry, and transitivity are not required.

   Solution Relate two elements exactly when they belong to the same one of the two given sets. Thus

   $$R = \{(p,p), (p,r), (r,p), (r,r), (q,q), (q,s), (s,q), (s,s)\}.$$

   The relation is reflexive because all four pairs (a, a) are listed. It is symmetric because each pair's reverse is also listed.

   For transitivity, suppose aRb and bRc. The elements a, b belong to one of the two given sets, and so do b, c. The sets are disjoint, so the set containing b is the same in both statements. Hence a, c belong to that same set, and aRc. Therefore R is an equivalence relation.

   By the listed pairs, $[p] = [r] = \{p, r\}$ and $[q] = [s] = \{q, s\}$. These are exactly the two required classes. ∎

Solutions revised using GPT-6 Astra in the Codex harness.